AP CALCULUS AB

Unit 1 • Practice • Answer key

Limits and Continuity

AP-style practice • Multiple choice and free response • Answer key with explanations

How to use this practice set

The questions follow the format of the AP Calculus AB exam. **Part A**: multiple choice, no calculator. **Part B**: multiple choice, graphing calculator required. **Part C**: free response; show your work and justify your answers, as on the exam.
Suggested timing: Part A 30 min • Part B 8 min • Part C 45 min.

Part A • Multiple choice • No calculator

Question 1 (1.6)

$\lim_{x\to 3} \dfrac{x^2-9}{x^2-2x-3}$ is

(A) 0
(B) $\frac{3}{2}$
(C) 3
(D) nonexistent

Solution

Answer: (B). $\dfrac{(x-3)(x+3)}{(x-3)(x+1)} = \dfrac{x+3}{x+1}$ for $x \neq 3$, and $\dfrac{6}{4} = \dfrac{3}{2}$.

Question 2 (1.6)

$\lim_{x\to 0} \dfrac{\sqrt{x+9}-3}{x}$ is

(A) 0
(B) $\frac{1}{3}$
(C) $\frac{1}{6}$
(D) 3

Solution

Answer: (C). Multiply by the conjugate: $\dfrac{(x+9)-9}{x\left(\sqrt{x+9}+3\right)} = \dfrac{1}{\sqrt{x+9}+3} \to \dfrac{1}{6}$.

Question 3 (1.8)

$\lim_{x\to 0} \dfrac{\sin(5x)}{x}$ is

(A) 5
(B) 1
(C) $\frac{1}{5}$
(D) 0

Solution

Answer: (A). $\dfrac{\sin 5x}{x} = 5 \cdot \dfrac{\sin 5x}{5x} \to 5 \cdot 1 = 5.$

Question 4 (1.15)

$\lim_{x\to\infty} \dfrac{4x^3 - x}{2x^3 + 7x^2}$ is

(A) 0
(B) $\frac{4}{7}$
(C) ∞
(D) 2

Solution
Answer: (D). Equal degrees: the limit is the ratio of the leading coefficients, $\frac{4}{2} = 2$.

Question 5 (1.15)

$\lim_{x\to-\infty} \dfrac{2x}{\sqrt{x^2+5}}$ is

(A) 2
(B) -2
(C) 0
(D) $-\infty$

Solution
Answer: (B). For $x < 0$, $\sqrt{x^2+5} \approx |x| = -x$, so the quotient approaches $\dfrac{2x}{-x} = -2$.

Question 6 (1.7)

$\lim_{x\to2^-} \dfrac{|x-2|}{x-2}$ is

(A) 1
(B) 0
(C) -1
(D) nonexistent

Solution
Answer: (C). For $x < 2$, $|x-2| = -(x-2)$, so the quotient equals -1.

Question 7 (1.4)

x	0.9	0.99	0.999	1.001	1.01	1.1
$f(x)$	2.71	2.97	2.997	3.003	3.03	3.31

Based on the table, which is the best estimate of $\lim_{x\to1} f(x)$?

(A) 2.997
(B) 3
(C) 3.003
(D) The limit does not exist.

Solution
Answer: (B). The values approach 3 from both the left and the right.

Question 8 (1.11)

Let $f(x) = \begin{cases} x^2+1, & x < 1 \\ 3x-1, & x \geq 1. \end{cases}$ Which statement is true?

(A) $\lim_{x\to1} f(x)$ does not exist.
(B) $\lim_{x\to1} f(x)$ exists, but f is not continuous at $x = 1$.
(C) f is continuous at $x = 1$.
(D) $f(1)$ is undefined.

Solution
Answer: (C). $\lim_{x\to 1^-} f(x) = 2$, $\lim_{x\to 1^+} f(x) = 2$ and $f(1) = 3 - 1 = 2$. All three conditions of continuity hold.

Question 9 (1.13)

For what value of k is $f(x) = \begin{cases} kx + 3, & x \le 2 \\ x^2 - k, & x > 2 \end{cases}$ continuous at $x = 2$?

(A) $-\frac{1}{3}$
(B) $\frac{1}{3}$
(C) 1
(D) 7

Solution
Answer: (B). Left: $2k + 3$. Right: $4 - k$. Then $2k + 3 = 4 - k$, so $3k = 1$ and $k = \frac{1}{3}$.

Question 10 (1.10)

What type of discontinuity does $g(x) = \dfrac{x^2 - 4}{x - 2}$ have at $x = 2$?

(A) Jump
(B) Infinite
(C) Removable
(D) None; g is continuous at $x = 2$.

Solution
Answer: (C). $g(2)$ is undefined, but $\lim_{x\to 2} g(x) = \lim_{x\to 2}(x + 2) = 4$ exists: a removable discontinuity.

Question 11 (1.14)

Which line or lines are vertical asymptotes of the graph of $h(x) = \dfrac{x + 1}{x^2 - 1}$?

(A) $x = 1$ only
(B) $x = -1$ only
(C) $x = 1$ and $x = -1$
(D) $y = 0$ only

Solution
Answer: (A). $h(x) = \dfrac{x + 1}{(x - 1)(x + 1)} = \dfrac{1}{x - 1}$ for $x \neq -1$. At $x = -1$ there is a hole (the limit is $-\frac{1}{2}$); at $x = 1$ the one-sided limits are infinite. $y = 0$ is a horizontal asymptote.

Question 12 (1.8)

$\lim_{x\to 0} x^2 \cos\left(\dfrac{1}{x}\right)$ is

(A) 1
(B) 0
(C) -1
(D) nonexistent

Solution
Answer: (B). $-x^2 \le x^2 \cos\frac{1}{x} \le x^2$ and both bounds approach 0. By the squeeze theorem, the limit is 0.

Question 13 (1.16)

The function f is continuous on $[1, 5]$ with $f(1) = -2$ and $f(5) = 6$. Which statement must be true?

(A) $f(c) = 8$ for some c in $(1, 5)$.
(B) $f(3) = 2$.
(C) f is increasing on $[1, 5]$.
(D) $f(c) = 4$ for some c in $(1, 5)$.

Solution
Answer: (D). 4 is between -2 and 6, and f is continuous, so the IVT guarantees such a c. Nothing forces $f(3) = 2$ or monotonicity, and 8 is not between the endpoint values.

Question 14 (1.15)

$\lim_{x \to \infty} \dfrac{3e^x + x^2}{e^x - 1}$ is

(A) 0
(B) 1
(C) 3
(D) ∞

Solution
Answer: (C). e^x dominates x^2 and constants as $x \to \infty$. Dividing by e^x: $\dfrac{3 + x^2 e^{-x}}{1 - e^{-x}} \to \dfrac{3+0}{1-0} = 3$.

Part B • Multiple choice • Graphing calculator

Question 15 (1.4)

Using a table of values or a graph, what is $\lim_{x \to 0} \dfrac{2^x - 1}{x}$, to three decimal places?

(A) 0.500
(B) 0.693
(C) 1.000
(D) 1.443

Solution
Answer: (B). Values at $x = \pm 0.001$ are about 0.6929 and 0.6934, so the limit is about 0.693 (it equals $\ln 2$).

Question 16 (1.16)

How many values of c in the open interval $(0, 2)$ satisfy $c^3 - 4c + 1 = 0$?

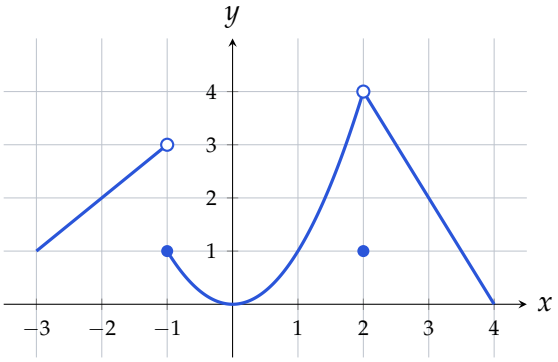
(A) None
(B) One
(C) Two
(D) Three

Solution
Answer: (C). Graphing $y = x^3 - 4x + 1$ shows zeros at $x \approx -2.115$, 0.254 and 1.861. Two of them are in $(0, 2)$.

Part C • Free response

Question 17 (Graph analysis • no calculator)

The graph of the function f on $[-3, 4]$ is shown. It consists of a line segment for $-3 \le x < -1$, the parabola $y = x^2$ for $-1 \le x < 2$, and a line segment for $2 < x \le 4$. Also $f(-1) = 1$ and $f(2) = 1$.



(a) Find $\lim_{x\to -1^-} f(x)$ and $\lim_{x\to -1^+} f(x)$. Does $\lim_{x\to -1} f(x)$ exist? Give a reason for your answer.
(b) Find $\lim_{x\to 2} f(x)$. Is f continuous at $x=2$? Name the type of discontinuity, if any, and explain how $f(2)$ could be redefined to make f continuous at $x=2$.
(c) Find $\lim_{x\to 3} f(f(x))$. Show the work that leads to your answer.
(d) $f(1)=1$ and $f(3)=2$. Explain why the Intermediate Value Theorem cannot be used to guarantee a value c in $[1,3]$ with $f(c)=1.5$. Does such a value exist? Justify.

Solution

(a) $\lim_{x\to -1^-} f(x)=3$ and $\lim_{x\to -1^+} f(x)=(-1)^2=1$. The one-sided limits are not equal, so $\lim_{x\to -1} f(x)$ does not exist.
(b) $\lim_{x\to 2^-} f(x)=2^2=4$ and $\lim_{x\to 2^+} f(x)=8-2(2)=4$, so $\lim_{x\to 2} f(x)=4$. But $f(2)=1\neq 4$, so f is not continuous at $x=2$. This is a removable discontinuity; redefining $f(2)=4$ makes f continuous at $x=2$.
(c) For x near 3, $f(x)=8-2x\to 2$. As $x\to 3^-$, $f(x)\to 2$ from above; as $x\to 3^+$, $f(x)\to 2$ from below. In both cases $f(f(x))$ approaches $\lim_{y\to 2} f(y)=4$ (from part (b)). So $\lim_{x\to 3} f(f(x))=4$. (Note that this is not $f(2)=1$.)
(d) The IVT requires f to be continuous on $[1,3]$, but f is not continuous at $x=2$. Such a value does exist: on $[1,2)$, $f(x)=x^2=1.5$ when $c=\sqrt{1.5}\approx 1.22$, which is in $[1,3]$.

Question 18 (Algebraic limits and continuity • no calculator)

Let $g(x)=\begin{cases}\dfrac{x^2+x-6}{x-2}, & x<2\\ cx+1, & x\geq 2\end{cases}$ where c is a constant, and let $h(x)=\dfrac{x^2+x-6}{x^2-4}$.

(a) Find $\lim_{x\to 2^-} g(x)$. Show the work that leads to your answer.
(b) Find the value of c for which g is continuous at $x=2$. Justify your answer using the definition of continuity.
(c) Find all values of x where h is not continuous, and classify each discontinuity. Justify with limits.
(d) Find $\lim_{x\to\infty} h(x)$ and $\lim_{x\to -2^+} h(x)$.

Solution

(a) For $x<2$: $\dfrac{(x+3)(x-2)}{x-2}=x+3$, so $\lim_{x\to 2^-} g(x)=5$.
(b) $g(2)=2c+1$ and $\lim_{x\to 2^+} g(x)=2c+1$. For continuity we need $\lim_{x\to 2^-} g(x)=\lim_{x\to 2^+} g(x)=g(2)$: $2c+1=5$, so $c=2$. Then $g(2)=5$ and $\lim_{x\to 2} g(x)=5=g(2)$.
(c) $h(x)=\dfrac{(x+3)(x-2)}{(x+2)(x-2)}$ is not defined at $x=2$ and $x=-2$. At $x=2$: $\lim_{x\to 2} h(x)=\lim_{x\to 2}\dfrac{x+3}{x+2}=\dfrac{5}{4}$ exists: removable discontinuity. At $x=-2$: the numerator of $\dfrac{x+3}{x+2}$ approaches 1 and the denominator approaches 0, so $\lim_{x\to -2^-} h(x)=-\infty$ and $\lim_{x\to -2^+} h(x)=$

$+\infty$: infinite discontinuity (vertical asymptote $x = -2$).
(d) $\lim_{x\to\infty} h(x) = 1$ (equal degrees, leading coefficients 1 and 1). $\lim_{x\to -2^+} h(x) = +\infty$.

Question 19 (Table and theorems • calculator)

The temperature W, in degrees Celsius, of a greenhouse is a continuous function of time t, in hours, for $0 \le t \le 10$. Selected values are given in the table.

t (hours)	0	3	5	8	10
$W(t)$ (°C)	20	24	31	27	22

(a) Must there be a time t in $[0, 10]$ at which $W(t) = 30$? Justify your answer.
(b) Explain why there must be at least two times in $[0, 10]$ at which $W(t) = 25$.
(c) Use the data to estimate the rate at which the temperature is changing at $t = 4$ hours. Show the computation and give units.
(d) A function f satisfies $4x - x^2 \le f(x) \le x^2 - 4x + 8$ for all x. Find $\lim_{x\to 2} f(x)$, and name the theorem you use.

Solution
(a) Yes. W is continuous on $[3, 5]$, $W(3) = 24$ and $W(5) = 31$. Since $24 < 30 < 31$, the IVT guarantees a t in $[3, 5]$ with $W(t) = 30$.
(b) W is continuous on $[3, 5]$ with $24 < 25 < 31$, so $W(t) = 25$ for some t in $[3, 5]$. W is continuous on $[8, 10]$ with $22 < 25 < 27$, so $W(t) = 25$ for some t in $[8, 10]$. These intervals do not overlap, so there are at least two such times.
(c) $\dfrac{W(5) - W(3)}{5 - 3} = \dfrac{31 - 24}{2} = 3.5$ degrees Celsius per hour.
(d) At $x = 2$: $4(2) - 2^2 = 4$ and $2^2 - 4(2) + 8 = 4$. Both bounding functions are polynomials, so $\lim_{x\to 2}(4x - x^2) = 4$ and $\lim_{x\to 2}(x^2 - 4x + 8) = 4$. By the squeeze theorem, $\lim_{x\to 2} f(x) = 4$.